

Hashmat Shadab Malik

PhD Candidate in Computer Vision | Multimodal Learning | Foundation Models | AI Safety

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Research Interests

My research focuses on **multimodal learning, foundation models, and AI safety**. I am particularly interested in vision-language models and multimodal large language models, trustworthy and reliable AI, adversarial robustness, generalization, and test-time adaptation, with applications spanning natural vision, medical imaging, and biometrics.

Education

- 2023–Present** **PhD in Computer Vision**, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE
GPA: **3.90/4.00**
- 2021–2022** **MSc in Computer Vision**, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE
Thesis: *Adversarial Pixel Restoration as a Pretext Task for Transferable Perturbations*
GPA: **4.00/4.00**
- 2014–2018** **BTech in Electronics and Communication Engineering**, National Institute of Technology Srinagar, India
Thesis: *Channel Estimation for Wireless Communication Systems Using the Least-Squares Method*
GPA: **8.48/10.00**

Research & Professional Experience

Graduate Research Assistant **Jan. 2021–Present**
Intelligent Visual Analytics Lab (IVAL), MBZUAI, Abu Dhabi, UAE

- Conduct research on multimodal learning and foundation models, with emphasis on AI safety, robustness, generalization, and reliable deployment of vision-language and multimodal models.
- Develop evaluation and defense methods for adversarial and distribution shifts, including test-time approaches, with applications to natural images, medical imaging, and biometrics.

Research Intern **Aug. 2025–Dec. 2025**
Robust Intelligent Systems, Khalifa University, Abu Dhabi, UAE

- Investigated multilingual adversarial robustness and multimodal safety of MLLMs, including cross-lingual attack transferability and safety failures under textual and visual jailbreak settings.

Research Intern **Mar. 2019–Jun. 2019**
Robotics Research Center, IIIT Hyderabad, India

- Developed deep learning methods for motion segmentation and multi-view depth estimation for autonomous navigation.

Computer Vision Engineer **Jul. 2018–Mar. 2019**
Cingularity TEC India Pvt. Ltd., Bangalore, India

- Built computer vision pipelines for automatic license plate recognition, vehicle recognition, and vehicle counting in parking and retail environments.

Project Assistant **Jul. 2018–Mar. 2019**
Computational Intelligence Lab, Indian Institute of Science (IISc), India

- Developed CNN-based systems for sugarcane disease recognition and vehicle damage classification/detection.

Selected Publications

- [1] **ReACT-CLIP: Response-Aware Test-Time Defense for Vision-Language Models.**
Hashmat Shadab Malik, Toluwani Aremu, Samuele Poppi, Muzammal Naseer, Salman Khan. *Under Review*.
[Paper]
- [2] **Investigating Adversarial Robustness of Multi-modal Large Language Models.**
Hashmat Shadab Malik, Muzammal Naseer, Salman Khan. *British Machine Vision Conference (BMVC), 2026*.
[Paper]
- [3] **Exploring Adversarial Robustness and Safety Alignment in Multilingual Multi-Modal Large Language Models.**
Hashmat Shadab Malik, Muzammal Naseer, Salman Khan. *British Machine Vision Conference (BMVC), 2026*.
[Paper]

- [4] **Beyond False Stability: High-Noise Drift Gating for Test-Time Adversarial Defenses in VLMs.**
Hashmat Shadab Malik, Muzammal Naseer, Salman Khan. *British Machine Vision Conference (BMVC)*, 2026. [Paper]
- [5] **CORTEX: A Structured Reasoning Benchmark for Trustworthy 3D Chest CT MLLMs.**
Hashmat Shadab Malik, Anees Ur Rehman Hashmi, Numan Saeed, Muzammal Naseer, Salman Khan, Christoph Lippert. *SAFER Workshop at MICCAI*, 2026. **Oral Presentation**. [Paper]
- [6] **How Good is my Histopathology Vision-Language Foundation Model? A Holistic Benchmark.**
Roba Al Majzoub, Hashmat Malik, Muzammal Naseer, Zaigham Zaheer, Tariq Mahmood, Salman Khan, Fahad Khan. *Medical Image Understanding and Analysis (MIUA)*, 2026. **Oral Presentation**. [Paper]
- [7] **FaceGuardian: Protecting Facial Biometrics from Malicious Generative Editing via Latent Optimization.**
Fahad Shamshad, Hashmat Shadab Malik, Muzammal Naseer, Salman Khan, Karthik Nandakumar. *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2026. **Oral + Poster; Best Paper Award at ECCV 2026 LifeGenIP Workshop**. [Paper]
- [8] **Robust-LLaVA: On the Effectiveness of Large-Scale Robust Image Encoders for Multi-modal Large Language Models.**
Hashmat Shadab Malik, Fahad Shamshad, Muzammal Naseer, Karthik Nandakumar, Fahad Khan, Salman Khan. *Trustworthy Foundation Models Workshop at ICCV*, 2025. **Oral Presentation + Poster**. [Paper]
- [9] **Hierarchical Self-Supervised Adversarial Training for Robust Vision Models in Histopathology.**
Hashmat Shadab Malik, Shahina Kunhimon, Muzammal Naseer, Fahad Shahbaz Khan, Salman Khan. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. [Paper]
- [10] **Towards Evaluating the Robustness of Visual State Space Models.**
Hashmat Shadab Malik, Fahad Shamshad, Muzammal Naseer, Karthik Nandakumar, Fahad Shahbaz Khan, Salman Khan. *Adversarial Machine Learning on Computer Vision Workshop at CVPR*, 2025. [Paper]
- [11] **ObjectCompose: Evaluating Resilience of Vision-Based Models on Object-to-Background Compositional Changes.**
Hashmat Shadab Malik, Muhammad Huzaifa, Muzammal Naseer, Salman Khan, Fahad Shahbaz Khan. *Asian Conference on Computer Vision (ACCV)*, 2024. **Oral Presentation; Best Student Paper Honorable Mention**. [Paper]
- [12] **On Evaluating Adversarial Robustness of Volumetric Medical Segmentation Models.**
Hashmat Shadab Malik, Numan Saeed, Asif Hanif, Muzammal Naseer, Mohammad Yaqub, Salman Khan, Fahad Shahbaz Khan. *British Machine Vision Conference (BMVC)*, 2024. [Paper]
- [13] **Adversarial Pixel Restoration as a Pretext Task for Transferable Perturbations.**
Hashmat Shadab Malik, Shahina K. Kunhimon, Muzammal Naseer, Salman Khan, Fahad Shahbaz Khan. *British Machine Vision Conference (BMVC)*, 2022. **Oral Presentation**. [Paper]

Honors & Awards

Sep. 2026	Best Paper Award , LifeGenIP Workshop at ECCV 2026, for <i>FaceGuardian</i> .
Dec. 2024	Best Student Paper Honorable Mention , Asian Conference on Computer Vision (ACCV 2024).
Jan. 2023	Four-year PhD Research Scholarship, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI).
Jan. 2021	Two-year Postgraduate Research Scholarship for MSc, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI).
Jun. 2018	Qualified the Graduate Aptitude Test in Engineering (GATE), India.

Academic Service

Reviewer: CVPR, ECCV, WACV, BMVC, MICCAI, ACCV.

Technical Skills

Programming	Python, C, MATLAB
Deep Learning	PyTorch, TensorFlow/Keras
Research Areas	Multimodal learning, foundation models, vision-language models, multimodal LLMs, adversarial robustness, AI safety, medical imaging

References

Dr. Salman Khan — Primary Advisor

Associate Professor, MBZUAI

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Dr. Fahad Shahbaz Khan — Secondary Advisor

Professor, MBZUAI

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Dr. Muzammal Naseer — Co-Supervisor

Assistant Professor, Khalifa University

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Dr. Numan Saeed — Research Collaborator

Assistant Teaching Professor & Research Scientist, MBZUAI

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